worldskills
Asia Taipei2025

Test Project

Mobile Applications Development

Module E Development and Testing

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Introduction

X-French Travel plans to develop several apps to help people learn about French tourist attractions, history, and culture and promote the flourishing development of local tourism.

This time, the App is designed to meet tourists' needs for booking hotels in the Alps. It provides comprehensive hotel information and booking services, simplifies the booking process, and improves users' travel experience in the Alps.

In this module, you need to develop the basic App functional logic and write automated test scripts to run the App.

Description of project and tasks

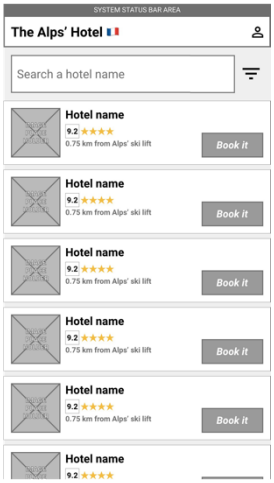
You should push a full project and an installable apk (Android) or app (iOS) package to Git Server.

Competition time: 3.0 hours

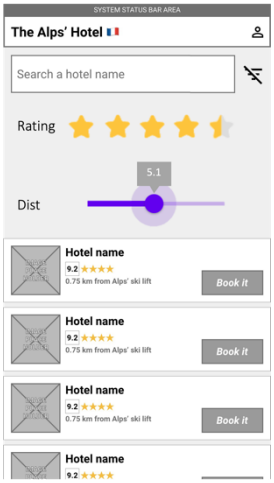
Marking Emulator:

- iOS: iPhone 16
- Android: Pixel 2

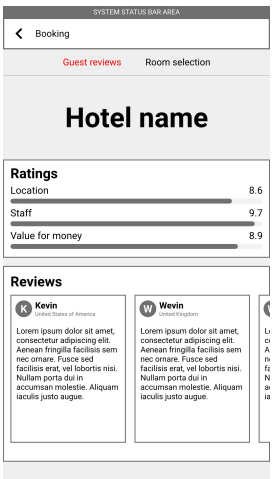
The App prototypes are for reference only. You do not need to copy the same appearance, colors, decorative icons/images, or elements' positions/style.



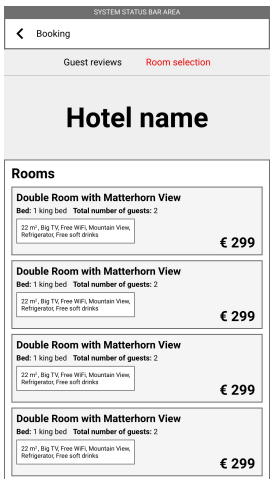
Home page



Home page(Filter)



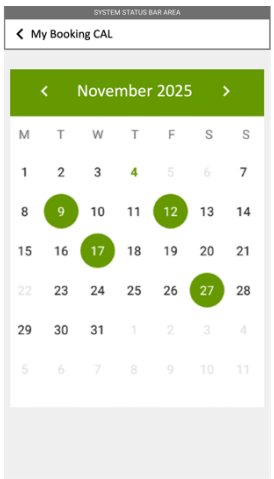
Booking Details page (guest reviews)



Booking Details page (room selection)

Booking Confirm page

My Bookings page



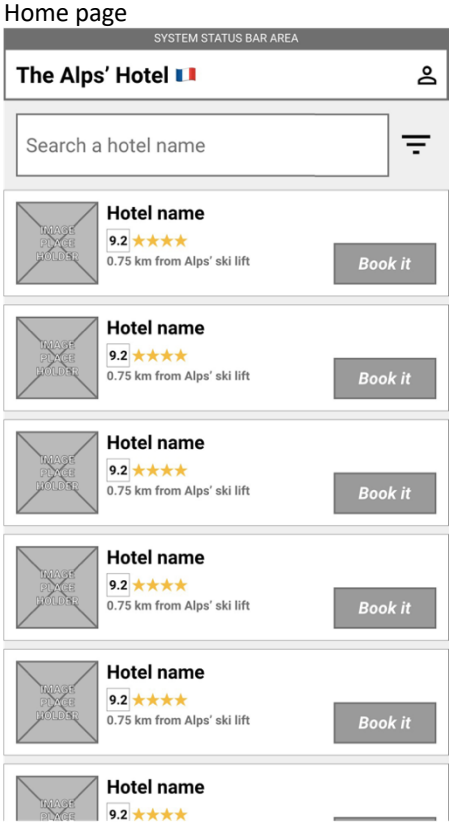
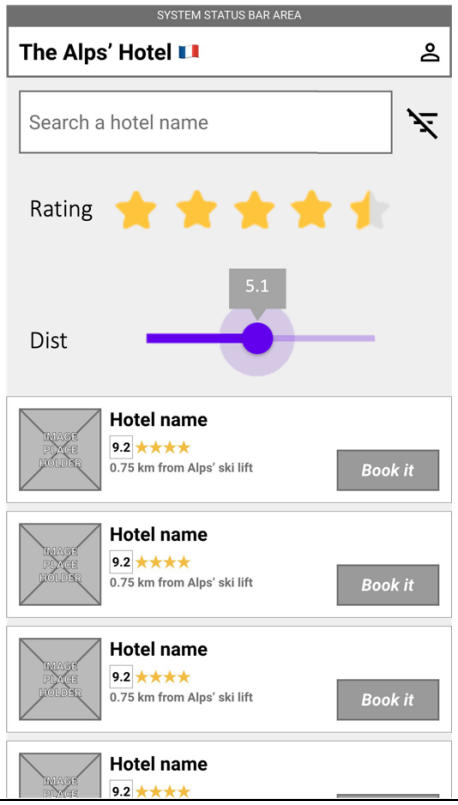
My Booking CAL page

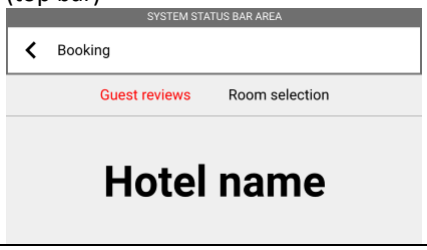
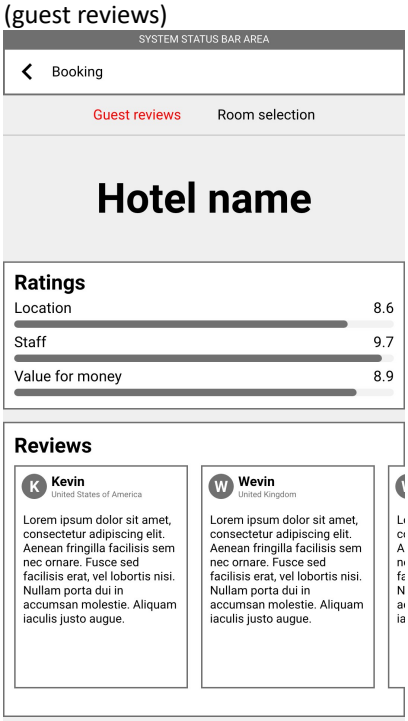
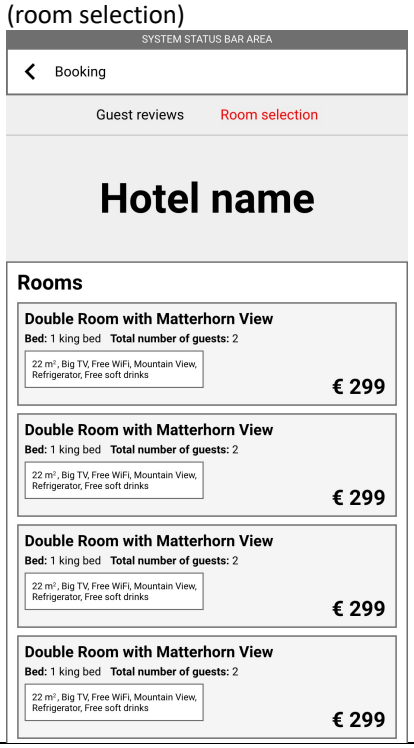
General Demands

1. All data you need are in **media-files/json-data**.
2. Your application functions should be developed according to all test cases (Form 1.1/1.2) and the automated function test case (Form 2.1).

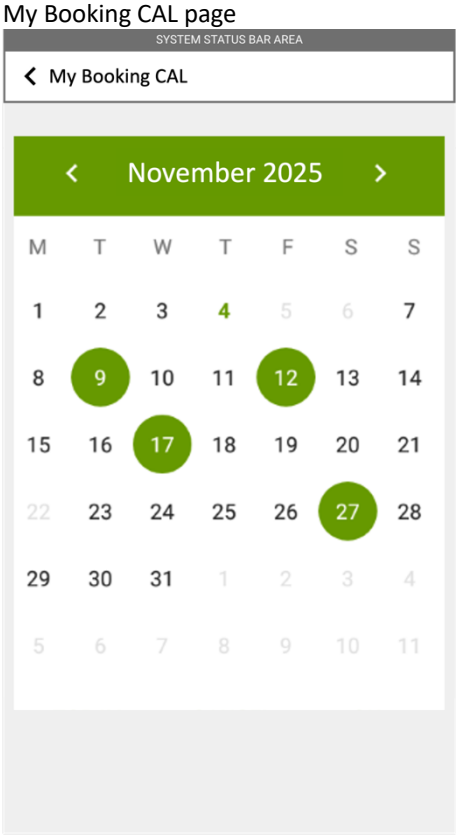
Develop by Test Case

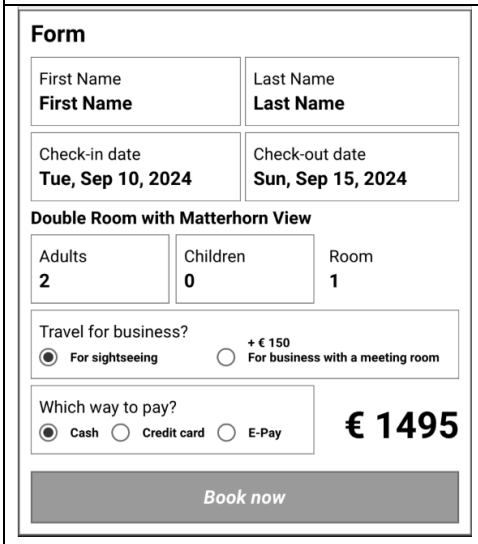
Develop the application by understanding the Test case.

FORM 1.1: APPLICATION UI TEST CASE		
No.	Area	Demands
1	Home page   	1. The title bar contains the app name, the French flag, and an icon button in the top right corner; 2. The homepage contains a search bar and a list of bookable hotels; 3. The search bar displays the hint text "Search a hotel name"; 4. The search bar allows finding hotels based on the entered text; 5. The Search button should be shown on the system keyboard; 6. A "Filter List" icon should be placed to the right of the search bar. When clicked, it should display filter options for "Rating" and "Dist". The original icon should change to a "Filter List Off" icon; 7. Rating should display five stars, and users must be able to select half-star increments. Dist should accept values from 1 to 10, with control down to one decimal place; 8. After applying filters, the search should be executed in the same way as described in point 5; 9. When the "Filter List Off" icon is selected, the Rating and Distance options should be hidden. In this case, searches should ignore Rating and Dist values, but any previously selected values should REMAIN saved; 10. Each hotel list item includes: (a) Hotel name; (b)Hotel ratings and star rating bars; (c) Distance-- "XX km from Alps' ski lift"; (d) Book Button; (e) Hotel cover image. 11. Click the booking button of a hotel list item to navigate to the corresponding Booking Details page (Only data with hotel_id 1000 and 1008 need to be implemented); 12. Click the icon button in the top right corner to navigate to the My Bookings page.

2	Booking Details (top bar)  	1. The title bar contains a back button and the text "Booking"; 2. Click the back button to navigate to the previous page; 3. The hotel information bar shows the hotel name; 4. Use tabs on the subpages (guest reviews, room selection) to navigate to the corresponding pages.
2.1	Booking Details page (guest reviews)  	1. The Ratings area includes: (a) Title--"Ratings"; (b) Several rating items; (c) Each rating item includes progress bars and scores. 2. The Reviews area includes: (a) Title--"Reviews"; (b) Horizontal list of reviews; (c) Each reviews list item includes name, nationality, circle shape and review content; (d) The initial in circle shape represents the first letter of the reviewer's name.
2.2	Booking Details page (room selection)  	1. The Rooms area includes: (a) Title--"Rooms"; (b) A list of rooms. 2. Each room includes: (a) Room Type; (b) Bed type-- "Bed: XX"; (c) "Total number of guests: XX"; (d) Room features introduction text; (e) "€" icon with price text. 3. Click a bookable room list item to navigate to the corresponding Booking Confirm page.

3	Booking Confirm page 	1. The title bar includes a back button, text "Booking Confirm", and the text "You are going to reserve:"; 2. Click the back button to return to the previous page; 3. Hotel information comes from previous selection; 4. The Form area includes: (a) "First Name" and "Last Name" input boxes; (b) "Check-in date" and "Check-out date" input boxes; (c) Room type to be booked; (d) "Adults" and "Children" input boxes; (e) Automatically calculated number of "Rooms"; (f) "Travel for business?" radio button with sub-options--"For sightseeing" and "+ €150 For business with a meeting room"; (g) "Which way to pay?" radio button with sub-options-- "Cash", "Credit card" and "E-Pay"; (h) "€" icon with automatically calculated price text; (i) "Book now" button.
4	My Bookings page 	1. The title bar includes the back button, and text "My Bookings"; 2. "List of my bookings" subtitle; 3. Booking List; 4. Each list item includes: (a) Numerical sequence (starting from 1); (b) Name of the person who made the reservation; (c) Start and end dates (formatted as: Tue, Sep 10, 2024); (d) Number of adults, children and rooms; (e) Booking type and payment method; (f) Price. 5. Sorting Controls (to the right of the subtitle, provide two sorting buttons): (a) "Calendar Clock" icon: • First click → sort by **Start Date** (descending); • Second click → sort by **Start Date** (ascending); • Third click → return to First click state. (b) "Attach Money" icon: • First click → sort by **Price** (descending); • Second click → sort by **Price** (ascending); • Third click → return to First click state. 6. "My Booking CAL" button.

5	My Booking CAL page  	1. The title bar includes the back button, and text "My Booking CAL"; 2. The calendar should highlight dates that have booking data based on the **Start Date**. For example, if there is a booking on **Fri, Nov 12, 2025**, that date should be emphasized on the calendar; 3. When a user clicks on a date with bookings in the calendar, the system should navigate back to the **My Bookings Page**, displaying **ONLY** the records with a **Start Date** matching the selected date; 4. The calendar must remain on the current system month. The text for today's date must be highlighted in a distinct color, clearly differentiated from booking dates, and must not match any other date.

FORM 1.2: APPLICATION TEST CASE					
No.	Area	Demands			
1	Search Area		Trigger Timing	Constraints Type	Constraints Description
		First Name Input and Last Name input	Click "Book now" Button	length	[1, 15]
				validity	Required, only letters can be entered
		Input box for the number of adults and children	Click "Book now" Button	length	Adults: [1, 5] Children: [0, 2 times the number of adults]
				validity	Required, cannot be children only, only numbers can be entered
				defaults	Adults: 1 Children: 0
		Date input box	Click "Book now" Button	format	(i.e. Tue, Sep 10, 2024)
				validity	Required, each value should be a valid date, and the start date should be earlier than the end date.

1. When the number of occupants changes, the number of rooms is automatically calculated based on the total number of people;
2. The price is automatically recalculated based on the number of rooms, the type of reservation and length of stay whenever the number of rooms, the reservation type or the date changes;
3. The date input box accepts multiple formats and automatically converts them to the correct format (e.g. "Tue, Sep 10, 2024"). Acceptable input formats only include "10/09/24", "2024-09-10" and "Sep 10 '24";
4. Clicking the "Book now" button triggers data verification.
   (a) If an error is detected, a corresponding prompt will pop up.
   (b) If the verification is successful, a prompt "Are you going to book this room?" will pop up. Clicking "No" closes the dialog box; clicking "Yes" saves the reservation information, close the dialog box, and navigates to the My Bookings page.

Automated Test

Finish the automated test script according to the test project description.
It is noted that the test script files must be stored in the location according to the **Automated Test Guide**.

All **STEPs** need to be held on for at least 10 seconds, and the "Step No: $Steps_No,$Description of the Action" should be logged on the "output" area, such as terminal, debug area or console.

Form 2.1: Application Function Test Case				
Program Name	The Alps' Hotel			
Test Type	Function Test			
Test Method	Black Box Test, Automated Test			
Test Demand	The workflow of the application functions runs normally.			
Test Execution Steps				
Steps_No.	Page	Description of the Action	Input Data	Expected Results
1		Application startup		Application starts normally
2	Home	Scroll to the bottom of the hotel list		The hotel list is successfully scrolled to the bottom
3	Home	Perform a search after entering the search term in the search bar	Meribel	Searching the hotel list results in the listing "Appartement Meribel"
4	Home	Click the listing item "Appartement Meribel"		Go to the Booking Details page for the hotel "Appartement Meribel" and display the Overview subpage
5	Booking Details (overviews)	Click the Guest reviews tab button		Display the Guest reviews subpage
6	Booking Details (guest reviews)	Scroll down the Reviews list to the 4th Review (published by "Miles")		Successfully scroll the Reviews list to the 4th Review item (publisher name is "Miles")
7	Booking Details (guest reviews)	Click the Room selection label button		Display the Room selection subpage
8	Booking Details (room selection)	Scroll down the room list to the "Great Family Room" item		Successfully scroll the room list to the "Great Family Room" item
9	Booking Details (room selection)	Click the "Great Family Room" item		Navigate to the Booking Confirm page of the hotel's Great Family Room
10	Booking Confirm	Enter First Name	Taylor	The First Name input box shows "Taylor"
11	Booking Confirm	Enter Last Name	Hutchinson	The Last Name input box displays "Hutchinson"
12	Booking Confirm	Enter check-in date	Dec 15 '25	The check-in date is displayed in the correct format
13	Booking Confirm	Enter check-out date	02/12/25	The check-out date is displayed in the correct format
14	Booking Confirm	Enter the number of adults	4	The number of adults is shown as 4 and the number of rooms is shown as 2
15	Booking Confirm	Enter the number of children	3	The number of children is shown as 3 and the number of rooms is shown as 3
16	Booking Confirm	Click the E-Pay radio button		The Cash button is disabled; E-Pay radio button is enabled
17	Booking Confirm	Click the "Book now" button		An error message pop-up window appears

18	Booking Confirm	Change check-out date	2025-12-19	The check-out date is displayed in the correct format
19	Booking Confirm	Click the "Book now" button		A confirmation pop-up window pops up to confirm the reservation
20	Booking Confirm	Click "Yes" in the pop-up window to confirm your reservation.		Navigate to the My Bookings page, and the booking list shows the information of the newly completed booking
21	Home	Click the back button to the Home page		Display the Home page
22	Home	Click top right corner icon		Navigate to the My Bookings page
23	Home	Click the back button to the Home page		Display the Home page
24	Home	Click the "Filter List" icon		Display Rating and Dist filters
25	Home	Select a Rating of 4.5 stars and a Dist of 6.9, then press the search key on the keyboard.		Display 'Residence Pierre & Vacances Premium les Crets' as matching the criteria
26	Home	Click the listing item "Residence Pierre & Vacances Premium les Crets"		Go to the Booking Details page for the hotel "Residence Pierre & Vacances Premium les Crets" and display the Overview subpage
27	Booking Details (guest reviews)	Click the Room selection label button & Scroll down the room list to the "Two-Bedroom Apartment - Open Mountain View" item & Click the item		Navigate to the Booking Confirm page
28	Booking Confirm	Enter check-in date & Enter check-out date & Enter the number of adults & Enter the number of children	Dec 21 '25 23/12/25 4 8	The number of rooms is shown as 2
29	Booking Confirm	Click For business with a meeting room radio button & Click the "Book now" button & Click "Yes" in the pop-up window		Navigate to the My Bookings page, and the booking list shows the information of the newly completed booking
30	My Bookings	Click the "Calendar Clock" icon		sort by **Start Date** (descending)
31	My Bookings	Click the "Calendar Clock" icon		sort by **Start Date** (ascending)
32	My Bookings	Click the "Calendar Clock" icon		sort by **Start Date** (descending)
33	My Bookings	Click the "Attach Money" icon		sort by **Price** (descending)
34	My Bookings	Click the "Attach Money" icon		sort by **Price** (ascending)
35	My Bookings	Click the "Attach Money" icon		sort by **Price** (descending)

36	My Bookings	Click the "My Booking CAL" button		Navigate to the My Booking CAL page
37	My Booking CAL	Click the control to move to the NEXT month in the calendar		Display two highlighted booking dates in October
38	My Booking CAL	Click December 21 to view booking records		Display booking details starting on December 21
39	My Bookings	Go back My Booking CAL		Display My Booking CAL
40	My Booking CAL	Click December 15 to view booking records		Display booking details starting on December 15

Instructions to the Competitor

1. Please create the project with correct package name (Android)/ Organization Identifier (iOS). Naming convention: org.wsa2025taipei.mexx ;
2. You should save the project in the folder: XX_Module_E;
3. You should rename the generated apk (Android)/ app (iOS) file as XX_Module_E.apk (Android)/ XX_Module_E.app (iOS), and save it in the root of the XX_Module_E folder;
4. The whole XX_Module_E folder should be pushed to the provided remote server in time;
5. Note: XX refers to your workstation code.